AGILE DEVOPS LAB FAT

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QUESTION:

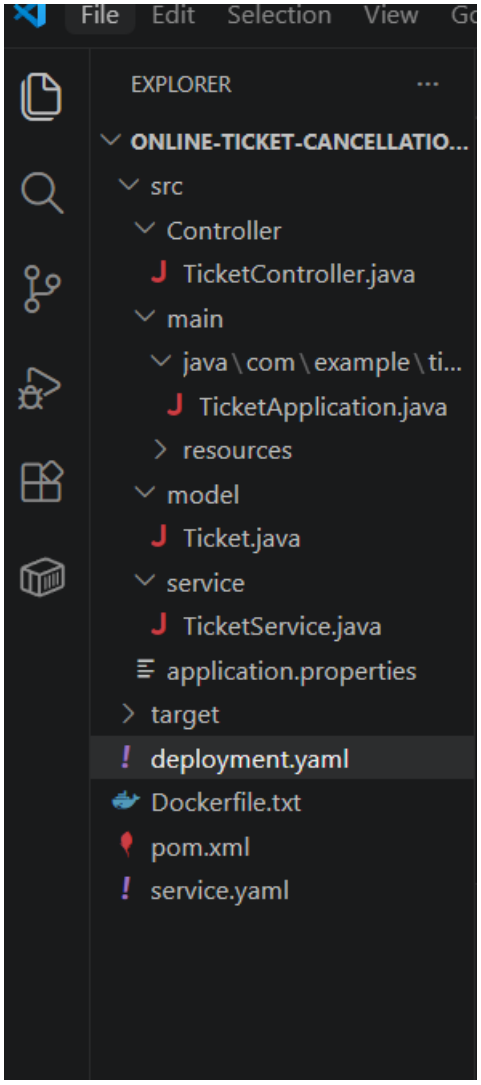
SET 3

ONLINE TICKET CANCELLATION SYSTEM

PROBLEM STATEMENT

Ensure tickets are cancelled and refunded correctly

ANS:



TicketController.java | TicketService.java | Ticket.java | application.p

src > Controller > TicketController.java

```java

import com.example.ticket.model.Ticket;
import com.example.ticket.service.TicketService;

@RestController
@RequestMapping("/tickets")
public class TicketController {

    private final TicketService service;

    public TicketController(TicketService service) {
        this.service = service;
    }

    // Book Ticket
    @PostMapping
    public Ticket bookTicket(@RequestBody Ticket ticket) {
        return service.bookTicket(ticket);
    }

    // Cancel Ticket
    @DeleteMapping("/{id}")
    public String cancelTicket(@PathVariable String id) {
        return service.cancelTicket(id);
    }

    // Get All Tickets
    @GetMapping
    public List<Ticket> getTickets() {
        return service.getAllTickets();
    }
}
```

TicketController.java | TicketService.java | Ticket.java | application.properties | pom.xml | Dockerfile.txt

src > service > TicketService.java

```java
import java.util.*;

import com.example.ticket.model.Ticket;

@Service
public class TicketService {

    private List<Ticket> tickets = new ArrayList<>();

    // Book Ticket
    public Ticket bookTicket(Ticket ticket) {
        ticket.setStatus("BOOKED");
        tickets.add(ticket);
        return ticket;
    }

    // Cancel Ticket
    public String cancelTicket(String ticketId) {
        for (Ticket t : tickets) {
            if (t.getTicketId().equals(ticketId)) {
                t.setStatus("CANCELLED");
                return "Ticket Cancelled Successfully";
            }
        }
        return "Ticket Not Found";
    }

    // Get All Tickets
    public List<Ticket> getAllTickets() {
        return tickets;
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

TicketController.java | TicketService.java | Ticket.java | application.properties | pom.xml

src > model > Ticket.java

```java
package com.example.ticket.model;

public class Ticket {

    private String ticketId;
    private String name;
    private String status;

    public String getTicketId() {
        return ticketId;
    }

    public void setTicketId(String ticketId) {
        this.ticketId = ticketId;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getStatus() {
        return status;
    }

    public void setStatus(String status) {
        this.status = status;
    }
}
```

TicketController.java | TicketService.java | Ticket.java

src > application.properties

```properties
server.port=8080
```

TicketController.java | TicketService.java | Ticket.java | application.properties | pom.xml | Dockerfile.txt | deployment.yaml

pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>ticket-system</artifactId>
    <version>1.0</version>

    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.2.0</version>
    </parent>

    <dependencies>
        <!-- Spring Boot Web -->
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <!-- Spring Boot Plugin -->
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
            </plugin>
        </plugins>
    </build>
```

TicketController.java | TicketService.java | Ticket.java | application.properties | pom.xml | Dockerfile.txt

Dockerfile.txt > ...

```dockerfile
FROM openjdk:17-jdk-slim
WORKDIR /app
COPY target/ticket-system-1.0.jar app.jar
ENTRYPOINT ["java", "-jar", "app.jar"]
```

TicketController.java | TicketService.java | Ticket.java | application.properties

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ticket-app
spec:
  replicas: 1
  selector:
    matchLabels:
      app: ticket-app
  template:
    metadata:
      labels:
        app: ticket-app
    spec:
      containers:
      - name: ticket-app
        image: 23mis0432/ticket-app:latest
        ports:
        - containerPort: 8080
```

TicketService.java | Ticket.java | application.properties | pom.xml | Doc

service.yaml

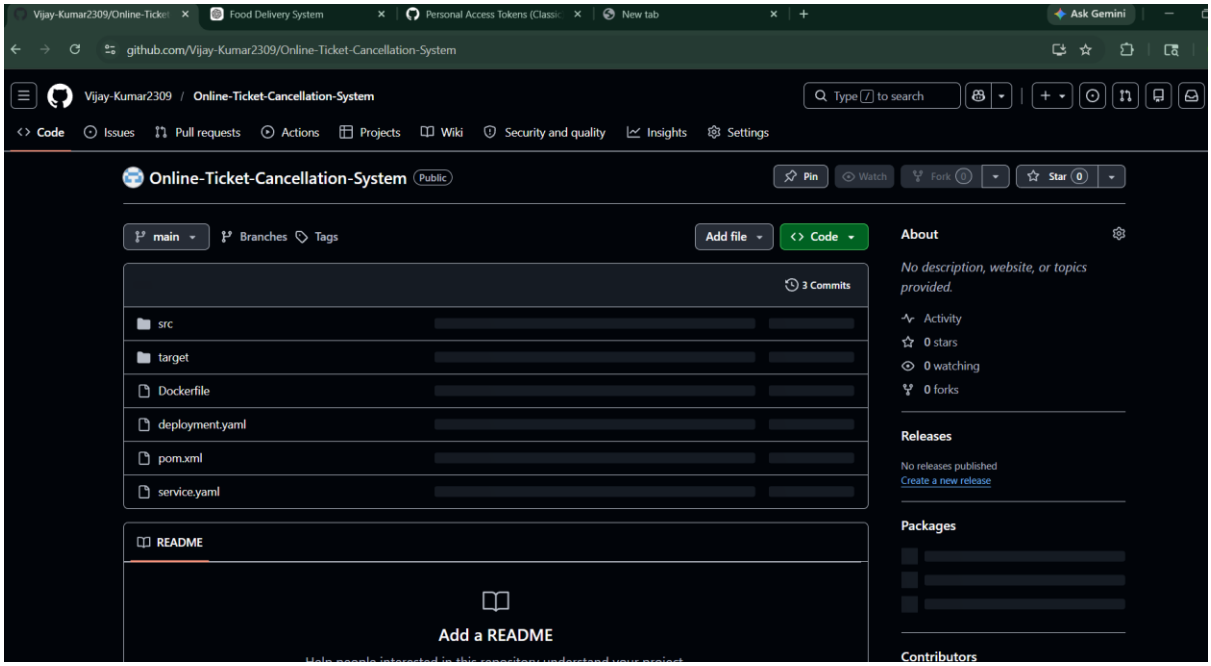
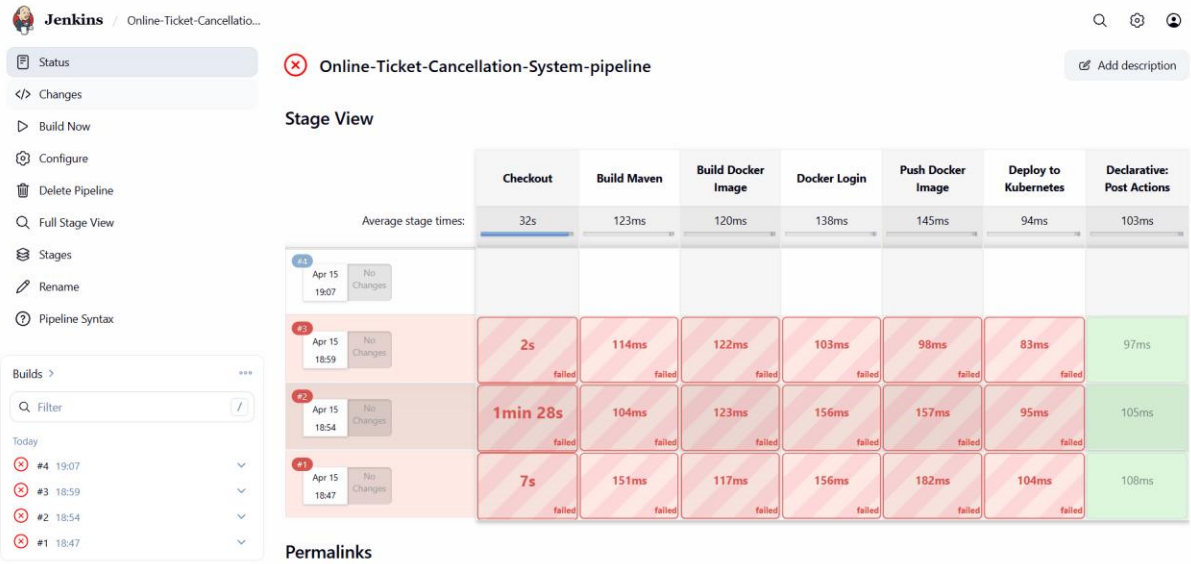
```yaml
apiVersion: v1
kind: Service
metadata:
  name: ticket-service
spec:
  type: NodePort
  selector:
    app: ticket-app
  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30008
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

[INFO]
[INFO] ----------------------< com.example:ticket-system >----------------------
[INFO] Building ticket-system 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.3.2:clean (default-clean) @ ticket-system ---
[INFO] Deleting C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ ticket-system ---
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ ticket-system ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug release 17] to target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ ticket-system ---
[INFO] skip non existing resourceDirectory C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ ticket-system ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ ticket-system ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ ticket-system ---
[INFO] Building jar: C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System\target\ticket-system-1.0.jar
[INFO]
[INFO] --- spring-boot:3.2.0:repackage (repackage) @ ticket-system ---
[INFO] Replacing main artifact C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System\target\ticket-system-1.0.jar w
ith repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System\target\ticke
t-system-1.0.jar.original
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System> git init
Initialized empty Git repository in C:/Users/Keerthi/OneDrive/Documents/Online-Ticket-Cancellation-System/.git/
PS C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be re
placed by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be repl
aced by CRLF the next time Git touches it
PS C:\Users\Keerthi\OneDrive\Documents\Online-T> git commit -m "first commit"
[master (root-commit) ecbae5a] first commit
 15 files changed, 186 insertions(+)
 create mode 100644 Dockerfile.txt
 create mode 100644 deployment.yaml
 create mode 100644 pom.xml
 create mode 100644 service.yaml
 create mode 100644 src/Controller/TicketController.java
 create mode 100644 src/application.properties
 create mode 100644 src/main/java/com/example/ticket/TicketApplication.java
 create mode 100644 src/model/Ticket.java
 create mode 100644 src/service/TicketService.java
 create mode 100644 target/classes/com/example/ticket/TicketApplication.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/ticket-system-1.0.jar
 create mode 100644 target/ticket-system-1.0.jar.original
PS C:\Users\Keerthi\On> git branch -M main           Cancellation-System>
PS C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System> git remote add origin https://github.com/Vijay-Kumar2309/
Online-Ticket-Cancellation-System.git
PS C:\Users\Keerthi\OneDrive\Documents\Online-Ticket-Cancellation-System> git push -u origin main
Enumerating objects: 36, done.
Counting objects: 100% (36/36), done.
Delta compression using up to 8 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (36/36), 16.87 MiB | 243.00 KiB/s, done.
Total 36 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Vijay-Kumar2309/Online-Ticket-Cancellation-System.git
 * [new branch]      main -> main
```



S	W	Name ↓	Last Success	Last Failure	Last Duration
✗	⛈	java-test-app-pipeline	1 day 21 hr #8	1 day 20 hr #10	1 min 2 sec
✗	⛈	java-test-app-pipeline1	N/A	6 hr 43 min #8	5.6 sec
✓	☁	javaci-cd	2 hr 25 min #18	2 hr 28 min #15	48 sec
✗	⛈	Online-Ticket-Cancellation-System-pipeline	N/A	3 min 1 sec #4	38 ms
⋯	☀	vnv	N/A	N/A	N/A